AD 2. AERODROMES

VOBZ AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VOBZ - VIJAYAWADA AIRPORT, VIJAYAWADA

VOBZ AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	163201N 0804812E MAG. BRG. 72.04 DEG/1498M from physical beginning of RWY 08	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	085 DEG, 23KM from Vijayawada City.	
3	Aerodrome elevation and reference temperature	83 FT / 40.3 DEG C	
4	Magnetic variation, date of information and annual change	1.25 DEG W (2010) /0.017 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	Airport Director Airports Authority of India Vijayawada Airport Gannavaram (Andhra Pradesh) - 521102,	
		Telephone:	(91) 8662846729 (91) 9441902480
		Fax:	(91) 866 28546757
		AFS:	VOBZYDYX VOBZYHYX
		Email:	apd_vobz@AAI.AERO
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	Elevation in EGM 08	

VOBZ AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	MON-FRI 0400-1230 UTC (0930-1800 IST) SAT, SUN+ HO: NIL
2	Custom and immigration	As ATS (See remarks No. 2)
3	Health and sanitation	Available within AD hours
4	AIS briefing office	As ATS
5	ATS reporting office (ARO)	As ATS
6	MET Briefing office	As ATS
7	Air Traffic Service	Consult NOTAM for Current ATS HR.
8	Fuelling	As ATS
9	Handling	As AD Administration.
10	Security	As ATS
11	De-icing	NIL
12	Remarks	(i) Outside of ATS hour's services are available O/R with 24 HR PN to AD. (ii) Customs and immigration facilities are available during flight schedule. 72-hour prior notice for any non-scheduled operations. (iii) The ATS approved hourly RWY traffic handling capacity as follows: Maximum number of arrivals and departures -12 Maximum number of arrivals only -06 Maximum number of departures only -10

VOBZ AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	Common User Domestic Cargo Terminal Available
2	Fuel and Oil types	JET A1
3	Fuelling facilities and capacity	HPCL (85KL), BPCL (75 KL) & IOCL (93 KL)
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	Nil
6	Repair facilities for visiting aircraft	Nil
7	Remarks	NIL

VOBZ AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	In the city
2	Restaurant(s) at or in the vicinity of aerodrome	At AD and in the city
3	Transportation possibilities	Buses, Taxis and Car hire AVBL
4	Medical Facilities	First aid at AD. Hospitals in the city.
5	Bank and post office at or in the vicinity of aerodrome	Banks: Nil Post office: Nil
6	Tourist office	A.P. Tourist Information counter AVBL in AD.
7	Remarks	NIL

VOBZ AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-7
2	Rescue equipment	AVBL as per category.
3	Capability for removal of disabled aircraft	1. The critical aircraft for the airport is A321. 2. Primary responsibility for removing disabled aircraft rests with the concerned airline. 3. The airlines may contact Airport Director for coordinating with the agencies for Disabled Aircraft Recovery Kit. 4. Contact Details of Airport Director: 0866-2846729, +91-9441902480 apd_vobz@aai.aero
4	Remarks	NIL

VOBZ AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	NIL
2	Clearance priorities	NIL
3	Remarks	NIL

VOBZ AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Refer Aircraft Parking/Docking Chart.
2	Designation, width, surface and strength of taxiways	Refer Aircraft Parking/Docking Chart Designator: TWY N
3	Location and elevation of altimeter checkpoints	Location:TWY A: 163147.21N 0804750.10E Elevation 76FT

4	Location of VOR checkpoints	TWY A, 163146.83 N 0804750.19 E, 78FT TXY B, 163144.84 N 0804740.73 E, 76FT TXY N, 163207.93 N 0804856.79 E 60FT TXY D
5	Position of INS checkpoints	NIL
6	Remarks	1. **Coordinates of RWY Holding Position** **TWY** **Coordinates** **Elevation (FT)** TWY A 163147.2167N 0804750.1079E 78 TWY B 163145.2179N 0804740.6442E 76 TWY D 163149.6302N 0804801.5150E 73 TWY N 163207.551N 0804856.8789E 60 2. **Coordinates of RWY & Taxi Intersection** **TWY** **Coordinates** **Elevation (FT)** TWY A 163150.4075N 0804749.3923E 80 TWY B 163148.4202N 0804739.9344E 76 TWY D 163152.7823N0804800.7811E 76 3. Dimension: Apron-I : 412 X 166.5 M Apron-II : 106 X 76 M Apron-III : 255.5 X 174 M 4. Isolation Bay: i. Location: North of RWY 8/26 at 524 m from beginning of Runway 26 ii. Dimension: 110 M X 120 M iii. Critical Aircraft: A321 iv. Coordinates: 163214.897 N 0804855.22 E v. Elevation: 60 ft vi. Parking Orientation: Facing North, Power-in/Power out vii. Surface and Strength: Concrete, 85/R/B/W/T 5. Elevation in EGM 08

VOBZ AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	Taxiing guidance provided on R/T, Guidelines at Apron
2	Runway and taxiway markings and lights	RWY Markings : Designation, THR, TDZ, Centreline, Edge, End RWY Lights : THR, Edge, End TWY Marking : Centreline, Holding Position & Edge TWY Lights : Edge on TWY A, TWY B, TWY N and TWY D RWY Guard on TWY D RWY Guard Lights : AVBL on TWY A,B,D and N
3	Stop bars (if any)	Nil
4	Remarks	NIL

VOBZ AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
26/TKOF 08/APCH	OTHER	163149.1N 0804720.9E	91 FT	NO	Mobile Road Traffic (Road Elev. 22.7m+5m Traffic Height)
26/TKOF 08/APCH	OTHER	163149.0N 0804719.6E	92 FT	NO	Mobile Road Traffic (Road Elev. 23.0m+5m Traffic Height)
26/TKOF 08/APCH	POLE	163148.4N 0804717.9E	107 FT	NO	Electric Pole
26/TKOF 08/APCH	POLE	163147.5N 0804716.7E	108 FT	NO	Electric Pole
26/TKOF 08/APCH	OTHER	163142.7N 0804715.5E	101 FT	LGTD	LC Rod on Localizer
26/TKOF 08/APCH	TREE	163144.8N 0804714.2E	102 FT	NO	Group of TreesGroup of Trees
26/TKOF 08/APCH	POLE	163137.8N 0804715.8E	100 FT	NO	Light Pole
26/TKOF 08/APCH	OTHER	163143.0N 0804714.1E	100 FT	NO	Mobile Road Traffic (Road Elev. 25.5m+5m Traffic Height)
26/TKOF 08/APCH	POLE	163144.3N 0804713.3E	106 FT	NO	Electric Pole
26/TKOF 08/APCH	TREE	163146.2N 0804712.0E	105 FT	NO	Group of Trees
26/TKOF 08/APCH	TREE	163142.4N 0804710.9E	109 FT	NO	Group of Trees
26/TKOF 08/APCH	TREE	163136.4N 0804711.9E	111 FT	NO	Group of Trees
26/TKOF 08/APCH	POLE	163136.1N 0804711.2E	119 FT	NO	Light Pole
26/TKOF 08/APCH	POLE	163136.8N 0804710.4E	118 FT	NO	Light Pole
26/TKOF 08/APCH	TREE	163137.8N 0804709.9E	113 FT	NO	Group of Trees
26/TKOF 08/APCH	TREE	163135.9N 0804708.6E	118 FT	NO	Group of Trees
26/TKOF 08/APCH	TREE	163133.5N 0804703.8E	125 FT	NO	Group of Trees
26/TKOF 08/APCH	TREE	163137.0N 0804659.7E	141 FT	NO	Group of Trees
26/TKOF 08/APCH	OTHER	163053.2N 0803922.9E	634 FT	NO	Hill Top
26/TKOF 08/APCH	SIGN	163136.3N 0804710.9E	111 FT	NO	Sign Board (International)
26/APCH 08/TKOF	OTHER	163214.0N 0804918.1E	68 FT	NO	Mobile Road Traffic (Road Elev. 15.7m+5m Traffic Height)
In circling area and at AD	BUILDING	163142.5N 0804739.9E	102 FT	NO	Customs Building

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	163141.7N 0804737.1E	157 FT	LGTD	Antenna On ATC (ATC Building Top - 42.4m)
In circling area and at AD	OTHER	163141.6N 0804737.1E	148 FT	LGTD	ATC Beacon Top (ATC Building Top - 42.4m)
In circling area and at AD	TREE	163138.6N 0804733.1E	150 FT	NO	Group of Trees
In circling area and at AD	TREE	163137.2N 0804722.7E	142 FT	NO	Group of Trees
In circling area and at AD	POLE	163149.3N 0804718.9E	106 FT	NO	Electric Pole
In circling area and at AD	TREE	163150.7N 0804717.9E	133 FT	NO	Group of Trees
In circling area and at AD	TREE	163136.7N 0804718.1E	133 FT	NO	Group of Trees
In circling area and at AD	TREE	163148.8N 0804714.0E	134 FT	NO	Group of Trees
In circling area and at AD	OTHER	163137.7N 0804716.3E	100 FT	NO	Mobile Road Traffic (Road Elev. 25.6m+5m Traffic Height)
In circling area and at AD	TREE	163134.7N 0804714.0E	151 FT	NO	Group of Trees
In circling area and at AD	POLE	163135.7N 0804712.1E	118 FT	NO	Light Pole
In circling area and at AD	TREE	163147.7N 0804655.9E	136 FT	NO	Group of Trees
In circling area and at AD	OTHER	163142.5N 0804734.8E	88 FT	LGTD	Wind Vane
In circling area and at AD	BUILDING	163209.9N 0804902.8E	64 FT	LGTD	New GP Hut Without Antenna
In circling area and at AD	OTHER	163209.9N 0804903.0E	102 FT	LGTD	LC Rod on GP Antenna (Antenna Top Elev. 30.3m)
In circling area and at AD	OTHER	163210.4N 0804905.5E	73 FT	LGTD	LC Rod on GP Monitor Antenna (Antenna Top Elev. 21.9m)
In circling area and at AD	OTHER	163210.4N 0804905.5E	72 FT	LGTD	New GP Monitor Antenna
In circling area and at AD	OTHER	163209.9N 0804903.1E	99 FT	LGTD	New GP Antenna
In circling area and at AD	OTHER	163210.0N 0804902.9E	74 FT	NO	New GP DME Antenna
In circling area and at AD	OTHER	163208.7N 0804916.8E	55 FT	NO	Approach Light
In circling area and at AD	TREE	163204.7N 0804927.7E	82 FT	NO	Group of Trees
In circling area and at AD	BUILDING	163202.3N 0804826.9E	76 FT	LGTD	GP Hut (Old)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	163204.5N 0804836.0E	80 FT	NO	Mobile Road Traffic (Road Elev. 19.3m+5m Traffic Height)
In circling area and at AD	OTHER	163202.3N 0804902.3E	80 FT	NO	WDI
In circling area and at AD	OTHER	163202.8N 0804905.9E	82 FT	NO	MET Antenna
In circling area and at AD	OTHER	163145.3N 0804743.8E	107 FT	NO	MET Antenna
In circling area and at AD	OTHER	163144.8N 0804745.1E	94 FT	LGTD	WDI
In circling area and at AD	TREE	163200.2N 0804801.0E	126 FT	NO	Group of Trees
In circling area and at AD	BUILDING	163201.4N 0804810.2E	106 FT	NO	Temple
In circling area and at AD	TREE	163201.6N 0804810.3E	140 FT	NO	Group of Trees
In circling area and at AD	TREE	163203.6N 0804820.2E	132 FT	NO	Group of Trees
In circling area and at AD	BUILDING	163202.8N 0804821.9E	92 FT	NO	Temple
In circling area and at AD	BUILDING	163203.9N 0804823.0E	105 FT	NO	Building
In circling area and at AD	TREE	163204.5N 0804823.6E	131 FT	NO	Group of Trees
In circling area and at AD	TREE	163204.0N 0804826.7E	119 FT	NO	Group of Trees
In circling area and at AD	BUILDING	163205.1N 0804826.9E	111 FT	NO	Building
In circling area and at AD	WALL	163203.2N 0804827.0E	78 FT	NO	Airport Boundary Wall with Fencing on Top
In circling area and at AD	BUILDING	163203.7N 0804827.7E	101 FT	NO	Pillar On Temple (Temple Top 29.5m)
In circling area and at AD	POLE	163204.2N 0804829.2E	90 FT	NO	Electric Pole
In circling area and at AD	TREE	163204.8N 0804829.5E	108 FT	NO	Group of Trees
In circling area and at AD	TREE	163206.9N 0804839.2E	103 FT	NO	Group of Trees
In circling area and at AD	BUILDING	163210.0N 0804844.0E	128 FT	NO	Building
In circling area and at AD	POLE	163208.1N 0804846.6E	86 FT	NO	Electric Pole
In circling area and at AD	TREE	163158.8N 0804852.5E	73 FT	NO	Group of Trees
In circling area and at AD	BUILDING	163206.6N 0804827.4E	124 FT	NO	Building
In circling area and at AD	BUILDING	163205.8N 0804824.8E	122 FT	NO	Building
In circling area and at AD	BUILDING	163204.0N 0804828.2E	85 FT	NO	Watch Tower

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	163231.8N 0804823.7E	251 FT	LGTD	Cellphone Mast
In circling area and at AD	OTHER	163722.0N 0804440.3E	655 FT	NO	Hill Top
In circling area and at AD	OTHER	163056.3N 0803926.2E	692 FT	LGTD	Mast on Hill
In circling area and at AD	OTHER	163045.7N 0803859.0E	920 FT	LGTD	Mast on Hill
In circling area and at AD	OTHER	163045.7N 0803859.2E	792 FT	NO	Hill Top
In circling area and at AD	OTHER	163045.3N 0803858.8E	812 FT	LGTD	Mast on Hill
In circling area and at AD	OTHER	163044.8N 0803859.4E	892 FT	LGTD	Mast on Hill
In circling area and at AD	OTHER	163925.1N 0804541.3E	710 FT	NO	Hill Top

VOBZ AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	AMO Vijayawada
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	Hyderabad 9 HR (00-09,03-12,06-15 and 09-18)
4	Availability of the trend forecast for the aerodrome and interval of issuance	NIL
5	Information on how briefing and/or consultation is provided	As ATS
6	Types of flight documentation supplied and language(s) used in flight documentation	Tabular form English
7	Charts and other information displayed or available for briefing or consultation	NIL
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	RVR Equipment and Digital Current Weather Instrument System (DCWIS) Equipment for RWY 26 And 08 available.
9	The air traffic services unit(s) provided with meteorological information	Vijayawada ATC and ACS
10	Additional information, e.g. concerning any limitation of service.	NIL

VOBZ AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCN) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
08	77.75 DEG	3360 x 45 M	90/F/B/W/T Asphalt	THR: 163146.32N 0804729.94E RWY END: 163208.28N 0804914.80E
26	257.75 DEG	3360 x 45 M	90/F/B/W/T Asphalt	THR: 163208.28N 0804914.80E RWY END: 163145.09N 0804724.08E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 80.7FT TDZ: 80.7FT	-0.26%	NIL	NIL	3480 x 280 M
THR: 54.8FT TDZ: 71.2FT	0.26%	NIL	NIL	3480 x 280 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
240M x 90M		---	1. RWY 08 PCN 0 to 1663 M 90/F/B/W/T 1663M to 2286M 99/F/B/W/T 2286M to 3360M 71/F/B/W/T 2. All elevations in EGM08
150M x 90M		---	1. RWY 26 PCN 0 to 1074 M 71/F/B/W/T 1074M to 1697M 99/F/B/W/T 1697M to 3360M 90/F/B/W/T 2. All elevations in EGM08

VOBZ AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
08	3360	3360	3360	3182	THR Displaced by 178M
26	3360	3360	3360	3360	

VOBZ AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
08	SALS 420 M LIH	Green	PAPI LEFT/3.00 DEG	
26	SALS 420 M LIH	Green	PAPI LEFT/3.00 DEG	

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
	3360 M 60 M White LIH	Red		NIL
	3360 M 60 M White LIH	Red		NIL

VOBZ AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	At Tower Building, FLG W&G EV2SEC As ATSAt Tower Building, FLG W&G EV2SEC
		IBN	NIL
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	255M East of ATS Tower,not lighted
		Anemometer	86M West of ATC Tower,not lighted
3	Taxiway edge and taxiway centre line lights;	Edge	AVBL for TWY ‘A,B, D and N’
		Centre Line	NIL
4	Secondary power supply including switch-over time;	Secondary Power supply to all lighting at AD. Switch-over time Zero sec (UPS available)	
5	Remarks	NIL	

VOBZ AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not established
2	TLOF and/or FATO area elevation:	Not established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not established
4	True bearings of FATO;	Not established
5	Declared distances available	Not established
6	Approach and FATO lighting;	Not established
7	Remarks	Not established

VOBZ AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Circular area centered on 163201N 0804812E within a 15NM radius.
2	Vertical limits	FL 65
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Vijayawada Tower, English
5	Transition altitude	4000 FT
6	Hours of applicability	HO
7	Remarks	Vijayawada Approach Control Unit is combined with Vijayawada Aerodrome Control Tower.

VOBZ AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	**2**	**3**	**4**
APP	Vijayawada Tower	118.550 MHZ	
APP	Vijayawada Tower	118.650 MHZ	
TWR	Vijayawada Tower	118.550 MHZ	
TWR	Vijayawada Tower	118.650 MHZ	
ATIS	Vijayawada Information	128.400 MHZ	
ALRS	EMERG FREQ	121.500 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	**6**	**7**
	HO	1. Main Frequency 2. Functions of Approach Control Service provided by Aerodrome Control Tower
		1. Backup Frequency 2. Functions of Approach Control Service provided by Aerodrome Control Tower
	As ATS	1. Main Frequency.

Logon address, as appropriate	Hours of operation	Remarks
	AS ATS	Alternate Frequency
	As ATS	NIL
	AS ATS	NIL

VOBZ AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 26 CAT I	IVJA	109.500 MHz	AS ATS
GP 26	IVJA	332.600 MHz	AS ATS
DME ILS 26	IVJA	CH32X	
DVOR/DME	BBZ	116.200 MHz CH109X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
163143.3N 0804715.5E			
163209.9N 0804903.1E			
163209.9N 0804903.1E	74 FT		1. Elevation in EGM08 2. Colocated with GP 26
163215.8N 0805008.2E	73 FT		Vertical datum: EGM08

VOBZ AD 2.20 LOCAL AERODROME REGULATIONS

1. PUSHBACK AND TAXI PROCEDURES

i. Aircraft parked on stands 2,3,4, can push back in both directions east or west and taxi out via taxiway D and B respectively. Aircraft parked on stands 14,15,16,17 shall pushback facing west only. Aircraft parked on apron III (stands 19 to 24) can pushback in both directions east or west and taxi out via taxiway D and B respectively.

ii. Simultaneous push back from apron III and any of the stands from 14 to 17 is permitted.

iii. Simultaneous push back from any of the stands from 2 to 4 and any of the stands from 14 to 17 is permitted.

iv. Simultaneous push back from apron III and any of the stands from 2 to 4 are not permitted.

v. Taxiway A and Apron II not available for Operations.

2. No Aircraft shall make lock turn/ pivotal turn on the Runway to avoid pavement damage.

LOW VISIBILITY PROCEDURES

1. INTRODUCTION:

Low Visibility Procedures (LVP) at Vijayawada Airport (VOBZ) are established to ensure the safe and efficient operation of aircraft during conditions of reduced visibility, in accordance with DGCA Civil Aviation Requirements, ICAO Annex 14, and the approved local Standard Operating Procedures.

Low Visibility Procedures at Vijayawada Airport (VOBZ) are applicable only to Category I (CAT I) operations and are established to facilitate take-off operations below the standard Category I take-off reference of 550 m RVR, as permitted under DGCA CAR Section 8, Series C, Part I. Due to the non-availability of runway centre line lights, take-off operations at this aerodrome are restricted to a minimum RVR of 400 m.

2. DEFINITIONS:

2.1 ILS Critical Area: An area of defined dimensions about the localizer and glide path antennas where aircraft and vehicles are excluded during all ILS operations. The critical area is protected because the presence of vehicles / or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.

2.2 ILS Sensitive Area: An area extending beyond the ILS critical area where the parking and/or movement of vehicles, including aircraft, are controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area Airports is protected to provide protection against interferences caused by large moving objects outside the critical area but still normally within the airfield boundary.

2.3 Low Visibility Procedures (LVP): Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

Note: as per para 5.3 of CAR on All Weather Operations, an operator shall not conduct Take-off with RVR/Visibility less than standard CAT-I conditions of 550m RVR/800m Visibility unless low visibility procedures are enforced.

2.4 Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

2.5 Movement Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the manoeuvring area and the apron(s).

2.6 Runway Visual Range: The range over which the pilot of an aircraft on the centre line of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

2.7 Safeguarding Procedures: Safeguarding Procedures (SP) are instructions for relevant airport departments and airside operators to prepare ground services and facilities for low visibility operations, in order that when LVP are implemented all SP are complete and airport is configured for Low Visibility Procedures and Low Visibility Take-offs.

2.8 Missed approach procedure: - The procedure to be followed if the approach cannot be continued.

2.9 Aerodrome Operating Minima- The limits of usability of an aerodrome for:

a) take-off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions.

b) landing in 2 D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and

c) landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation.

2.10 Touchdown Zone- The portion of a runway, beyond the threshold, where it is intended landing aeroplanes first contact the runway.

2.11 Visibility- Visibility for aeronautical purposes is the greater of:

a) the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background.

b) The greatest distance at which lights in the vicinity of 1000 candelas can be seen and identified against an unlit background.

Note 1. The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

Note 2. The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

3. ABBREVIATIONS:

ADC: Aerodrome Control
ARFF: Airport Rescue and Fire Fighting Services
ATC: Air Traffic Control
ATM: Air Traffic Management
CFT: Crash Fire Tender
ATIS: Automatic Terminal Information Service
DG: Diesel Generating Set
LVP: Low Visibility Procedures
MET: Meteorology
OPS: Operations
RWY: Runway
RVR: Runway Visual Range
SMC: Surface Movement Control
SP: Safeguarding Procedures
TDZ: Touchdown Zone
TMO: Tower Metrological Officer
TWY: Taxiway

4. LOW VISIBILITY PROCEDURES:

4.1 Regulations: CIVIL AVIATION REQUIREMENT SECTION 8 - AIRCRAFT OPERATIONS SERIES 'C' PART I ISSUE II, DATED 28TH OCTOBER 2022:

An operator shall not conduct take-off with RVR/visibility less than standard Category I conditions of 550m RVR/800 m visibility unless low visibility procedures are enforced.

An operator shall not conduct low visibility take-offs in less than 400 m RVR unless approved by DGCA. Operators may be authorized LVTO minima of up to 75m. Foreign operators, who are authorized by their State Regulatory Authority for LVTO, shall submit requisite documents to DGCA for approval of LVTO at Indian aerodromes.

5. GENERAL:

5.1 Low visibility procedures have been developed for Vijayawada Airport, Vijayawada to accommodate/permit departures in visibility/RVR less than 800M/550M from RWY 26 beginning (runway served with touchdown zone RVR instruments).

5.2 The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities.

6. MINIMUM REQUIREMENTS:

6.1 The following Aeronautical Ground lights and RVR equipment shall be serviceable to the required standards to support Low Visibility Procedures:

a) Runway edge lights,
b) Runway end lights,
c) Runway Guard Lights (available on TWY 'A', 'B', 'D' and 'N')
d) Instrument RVR (Available for TDZ Runway 08/26).
e) Stand by Power supply to maintain switch over time of 1 Second for Runway

Edge Lights and Runway End Lights. This requirement can be met with the help of DG Set and/or UPS.

6.2 Unserviceability of Aeronautical Ground Lights/ Equipment before Implementation of LVP.

6.2.1 Low Visibility Procedures will not be implemented when any of the light/equipment mentioned in para 6 above is un-serviceable or is not maintained as per the required standard.

Aeronautical Ground Lighting Facility	Un-serviceability	Restrictions
Runway Edge lights	More than 5% of all lights are unserviceable	LVP operations will be suspended.
	Any two consecutive lights or more are unserviceable	
Runway End lights	More than 15% of all lights are unserviceable	LVP operations will be suspended.
	Any two consecutive lights or more are unserviceable	
Standby Generators/UPS	Any of the generator/UPS is unserviceable	LVP operations will be suspended.

Note: When any of the light/equipment mentioned in para 6 above becomes un- serviceable or fails to meet the required standard during periods of LVP, Tower shall advise the aircraft accordingly and LVP shall be suspended and information to this effect shall be included in ATIS broadcast.

6.3 Runway 08/26 beginning (full length) only will be used for departures under LVP.

6.4 Only one aircraft movement at a time shall be permitted in the movement area when the RVR is less than 550 metres.

6.5 Due to the non-availability of Runway Centre Line lights, take-off operations are restricted to a minimum RVR of 400 m.

7. EQUIPMENT AND SERVICES

7.1 Equipment

a) DCWIS (Digital Current Weather Indication System) at Runway 08/26.

b) FSM (Forward Scatter Meter) at Runway 08/26.

c) RVR is reported in the following scales:

i) The maximum reportable value of RVR is 2000 m. When RVR is above 1500 m, it is reported as 2000 m.

ii) In the increments 100 m when greater than 800 m.

iii) In the increments 50 m when RVR greater than 400 m but less than 800 m.

iv) In the increments 25 m when less than 400 m.

d) In case of unserviceability of the above metrological equipment's, the LVP shall be suspended.

7.2 ILS

7.2.1 Runway 26 is equipped with CAT I ILS.

8. SAFEGUARDING PROCEDURES

8.1 Safeguarding Procedures (SP) are instructions for relevant airport, departments and airside operators to prepare ground services and facilities for low visibility operations in order that when LVP are implemented all Safeguarding procedures are complete.

8.2 Safeguarding Procedures shall be initiated when-

a) The Visibility/RVR is less than 1200m and visibility/RVR is forecast to deteriorate to 800m or less; and/or

b) The cloud ceiling is less than 400ft and forecast to fall to 200ft or less.

8.3 Safeguarding procedures include-

a) Positioning of CFTs as per Visibility Standby procedures.

b) Stopping of all maintenance works on the manoeuvring area, ILS sensitive and critical area as removal of all men and mobile equipment from the said area.

c) Ensuring availability of secondary power supply for change over time of maximum one second for Runway Edge and Runway End lights supported by UPS.

d) The appropriate Aeronautical ground lights must have been inspected during the hour preceding implementation of LVP, and thereafter once every two-hour period. These lighting inspections should be accorded priority and, if necessary, aircraft operations may be delayed.

e) The mandatory serviceability requirements of the Aerodrome Ground Lightings (AGLs) system shall be maintained as per CAR Section 4, Series B, Part I, Para 10.5.11, through inspection by Follow-Me with Electrical personal and reporting to ATCO prior to each departure when the RVR is less than 550 metres.

Note- Runway Edge and Runway End lights may continue to operate on main power supply during safeguarding Procedures. Whenever, LVP is to be implemented as per para 9 below, the Runway Edge and Runway End lights shall be put on Standby Power Supply (DG set or UPS). This operation needs to be completed before LVP is implemented. In case of DG set, Main power supply shall act as stand by power. If UPS is capable of maintaining the required AGL system with one second of Switch Over time with Main Supply, the main supply can continue to be primary supply and the Generator Supply can be kept as Stand by Power supply.

f) Operations department shall ensure that all runway access roads are closed.

8.4 Process of initiation of Safeguarding Procedures-

a) When meteorological conditions meet the criteria for initiation of SP as stipulated in Para 8.2, the Tower Supervisor/Duty Officer will initiate and co- coordinate with CCR, SSO (CNS), Duty Manager (Airside) and Fire Station for completion of safeguarding procedures before implementation of Low Visibility Procedures.

b) When all the CCR, SSO (CNS), Duty Manager (Airside) and Fire Station have completed their necessary actions, they shall report to Tower Supervisor/Duty Officer that their Safeguarding Procedure (SP) is completed, and the airport is safeguarded for LVP operations.

9. LOW VISIBILITY OPERATIONS

Low Visibility Procedures are the procedures to ensure the safe operation of aircraft during periods of reduced visibility or low cloud base. LVP shall only be implemented when Safeguarding Procedure (SP) has been completed, and the airport is configured for low visibility operations.

9.1 Implementation of Low Visibility Procedures-

a) Whenever Visibility/RVR reduces to 800 Meters or below and/ or cloud ceiling is at 200 ft or below the Tower Supervisor/ Duty Officer shall coordinate with CCR, SSO (CNS), Duty Manager (Airside) and Fire Station to confirm whether the Safeguarding procedures have been completed or not.

b) Once, it is confirmed that Safeguarding Procedures are implemented and LVP is necessitated, the Tower Supervisor/Duty Officer will implement Low Visibility Procedures and shall inform CCR, SSO (CNS), Duty Manager (Airside) and Fire Station of the imposition of low visibility procedures.

9.2 Actions by Tower Controller

a) Whenever visibility/RVR is less than 800M/550M, Tower shall confirm from pilot that the reported RVR value is within minima before issuing take-off clearance.

b) Tower shall ensure that the towing of aircraft is done under escort of "Follow Me" vehicles. "Follow Me" shall inform ATC of any deviation from the route cleared by ATC.

c) Tower shall not permit any high-power ground run on the manoeuvring area except idle power run on the stands.

d) In case high power engine run-up is in progress, the same shall be stopped whenever RVR reduces below 550m and towed back to respective stand.

e) Tower shall ensure that "Follow Me" services are provided to pilots on request.

f) The number of vehicles on the manoeuvring area shall be restricted to bare minimum and records of all operating on the manoeuvring area shall be maintained by Tower.

g) Check ATIS broadcast and include the message that "Low Visibility Procedures in Force". If ATIS is not available inform aircrafts on VHF.

9.3 Actions by SSO

9.3.1 On receipt of "Initiating SP" from Tower Supervisor/Duty Officer, SSO will inform the Duty Officer, NAV AIDS and have the ILS equipment and its status indicators in ATC units checked up.

9.3.2 On receipt of "Initiating LVP" from Tower Supervisor/Duty Officer that LVP are to be made effective SSO will maintain continuous watch on the performance of ILS equipment.

9.3.3 He will inform Tower Supervisor/Duty Officer of any unserviceability in the equipment which is likely to affect Low Visibility operations.

9.4 Actions by Duty Manager (Airside)

9.4.1 On receipt of advice from Tower Supervisor/Duty Officer to implement Safeguarding procedure, the Terminal Duty Manager, will immediately inform the following:

i) CASO (State Office)

ii) All airlines/GHA.

iii) Civil In-charge

9.4.2 Duty Manager (Airside) shall ensure that-

9.4.2.1 No vehicle/person enters or is present in the sensitive/critical areas of localizer and glide path.

9.4.2.2 All civil/electrical works in progress are to be stopped in the manoeuvring area immediately and men/material/ equipment to be removed from the sensitive/critical areas of localizer and glide path.

9.4.3 Duty Manager (Airside) would advise Tower Supervisor/Duty Officer when all actions allocated to them are completed.

9.4.4 Subsequently, Duty Manager (Airside) shall remain available in apron and will maintain listening watch on Walkie-Talkie.

9.4.5 The vehicles which are not equipped with RT but has to enter the runway or taxiway for urgent operational requirement shall be escorted all the time.

9.4.6 Duty Manager (Airside) shall ensure that all the roads are closed during the Low Visibility Operations, except the Fire approach road and designated PDPs.

9.5 Actions by Shift In-Charge (Civil)

9.5.1 Shift In-Charge (Civil) to ensure that all civil works in progress in manoeuvring area, are stopped and that the work area is restored in complete serviceable condition and confirm to Tower Supervisor/Duty Officer accordingly.

9.5.2 Shift In-Charge (Civil) to ensure that during Low Visibility Operations, no equipment, manpower shall be present in sensitive areas of localizer and glide path.

9.5.3 Shift In-Charge (Civil) to ensure that none of the civil contract workers/vehicles shall enter the manoeuvring area during the Low Visibility operations.

9.6 Actions by Shift In-Charge (Electrical)

9.6.1 On receipt of advice to implement Low Visibility Procedures from Tower Supervisor/Duty Officer, will check that following visual aids are serviceable and can be operated at full intensity:

a) Approach lighting system

b) Runway edge lights

c) Runway threshold and end lights

d)Runway guard lights

e)Taxiway edge lights

(NOTE - No adjustment in light intensity shall be made without permission from ATC Tower during LVP)

9.6.2 Shift In-Charge (Electrical) shall ensure that no electrical maintenance works is carried out during LVP either in powerhouse or on any other electrical facilities used during Low visibility operations.

9.6.3 Shift In-Charge (Electrical) will ensure that Runway switch room is manned and position themselves at halls for ground lighting facilities.

9.6.4 Shift In-Charge (Electrical) will inform the un-serviceability or any change in status of any facility/systems to Tower Supervisor/Duty Officer immediately.

9.6.5 Shift In-Charge (Electrical) shall ensure the electrical personal is available in follow me vehicle for the AGL inspections before each departure during LVP implementation.

9.7 Actions by ARFF

9.7.1 Positioning CFTs as per visibility standby procedures.

9.7.2 ARFF/ CFTs must obtain clearances from Tower prior to entry to movement area.

9.8 Actions by Tower Met Officer (Meteorological Office)

9.8.1 Tower Met Officer would issue an 'Outlook for Low Visibility Procedures' to the Tower Supervisor/Duty Officer of air traffic services whenever he expects that the RVR and/or cloud ceiling will fall below 800 m and/or 200ft or less respectively.

9.8.2 Whenever Tower Met Officer visualizes that RVR is likely to fall below 800 m and/or cloud ceiling to 200ft or less within next 2 hours, he will inform to Tower Supervisor/Duty Officer to this effect.

9.8.3 When the RVR and/or cloud ceiling are 800m and/or 200ft respectively and the trend is towards improvement in these elements of weather conditions the Duty Met Officer may, when requested by Tower Supervisor/Duty Officer, advise him about such improving weather conditions for the purpose of termination of LVP.

9.8.4 The Tower Met Officer shall be present in Tower during periods of reduced visibility and ensure that the RVR displays in ATC units in the Tower and Approach Control are serviceable.

10. TERMINATION OF LOW VISIBILITY PROCEDURES

10.1 When Visibility/RVR improves to 800M or more and cloud ceiling is 200 feet or Higher and trend is for improvement, Tower Supervisor/Duty Officer Tower would terminate operations of LVP. He may obtain advice from Tower Met Officer regarding improvement in weather conditions before the termination of LVP.

10.2 On termination of LVP, following message shall be included in two subsequent ATIS broadcasts. “LOW VISIBILITY PROCEDURES TERMINATED”. If ATIS is not available inform aircrafts on VHF.

10.3 If SP are implemented and LVP are not subsequently implemented and the visibility/RVR improves and is more than 1200 m and/or the cloud ceiling is 400ft or higher and both are forecast to remain above the required SP criteria, Tower Supervisor/ Duty Officer Tower may cancel SP.

11.OTHER ACTIONS

11.1 Every year before commencement of monsoon/winter season, a meeting will be held by Airport Director, to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure co-operation for safe airport operations during periods of low visibility.

11.2 Before issuing the Airside driving permits, the OPS Incharge shall ensure that staff and drivers are suitably trained for Low Visibility operations.

11.3 All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

11.4 All the vehicles must have their obstruction lights “ON” during Low Visibility Procedures operations.

11.5 A refresher program for ATCO’s and personnel responsible for airside operations shall be conducted every year.

VOBZ AD 2.21 NOISE ABATEMENT PROCEDURES

NIL

VOBZ AD 2.22 FLIGHT PROCEDURES

NIL

VOBZ AD 2.23 ADDITIONAL INFORMATION

1. Take-off Run Available for Departures from Intermediate RWY Positions:
   i. RWY 08:
      a) From TWY A – 2589 M
      b) From TWY B – 2867 M
      c) From TWY D – 2240 M
   ii. RWY 26:
      a) From intermediate turn pad: 2286 M
2. Departure on RWY 26 from intermediate turn pad can be initiated by ATC with concurrence of pilot or as per request from pilot.
3. Aircraft landing on RWY 08 can use intermediate turn pad for making 180 degree turn for runway vacation. Critical aircraft A321.
4. ADS-B Commissioned with following Details:
   i. FREQ: 1090MHz
   ii. COORD: 16.528826N 80.793660E
   iii. SITE ELEV: 61.728216 M/202.53 ft AMSL
5. Aircraft stands 15 and 16 are not available for night operations.

VOBZ AD 2.24 CHARTS RELATED TO AN AERODROME

1. Aerodrome Chart (02 OCT 2025)
2. Aircraft Parking/Docking Chart (02 OCT 2025)
3. Aerodrome Obstacle Chart-Type A (Operating Limitations) RWY 08 (12 JUN 2025)
4. Aerodrome Obstacle Chart-Type A (Operating Limitations) RWY 26 (12 JUN 2025)
5. Aerodrome Obstacle Chart-Type B (12 JUN 2025)
6. Coding Table - ILS (Y) Procedure RWY 26
7. VOR Procedure RWY 08
8. VOR Procedure RWY 26
9. RNP Procedure RWY 08
10. Coding Table - RNP Procedure RWY 08
11. RNP Procedure RWY 26
12. Coding Table - RNP Procedure RWY 26

VOBZ AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	**2**	**3**